Small Office Network with DHCP, DNS, and Intranet

Platform: Cisco Packet Tracer

Project Type: Network simulation

Objective

Design and test a small office network with separate Admin and IT departments. Configure automatic IP addressing, allow communication between the departments, and provide an internal website that employees can access using a domain name.

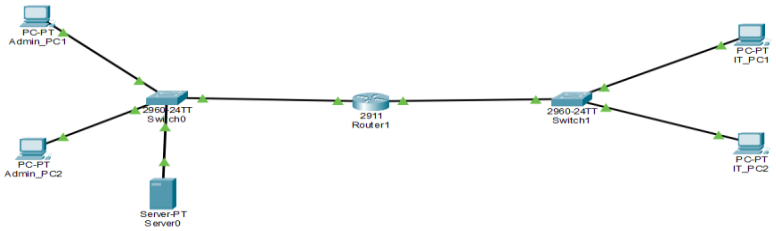
Network Design

The simulated office contains:

- One router
- Two switches
- Four PCs: two Admin and two IT
- One server providing DNS and HTTP services

The network is divided into two subnets:

Department	Network	Default gateway
Admin	192.168.10.0/24	192.168.10.1
IT	192.168.20.0/24	192.168.20.1



Configuration

Router

The router was configured with an interface in each subnet:

- Admin interface: 192.168.10.1/24
- IT interface: 192.168.20.1/24

DHCP pools were configured on the router so that PCs could receive their IP configuration automatically.

DHCP pool	Network	Gateway	DNS server
ADMIN	192.168.10.0/24	192.168.10.1	192.168.10.2
IT	192.168.20.0/24	192.168.20.1	192.168.10.2

Server

The server was assigned a static IP address:

Setting	Value
IP address	192.168.10.2
Subnet mask	255.255.255.0
Default gateway	192.168.10.1

The server's DNS service was configured with the record:

intranet.office → 192.168.10.2

The HTTP service was enabled to host the internal website.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip helper-address 192.168.10.2
                  ^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.99
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.99
Router(config)#
Router(config)#ip dhcp pool ADMIN
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.10.2
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 192.168.10.2
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

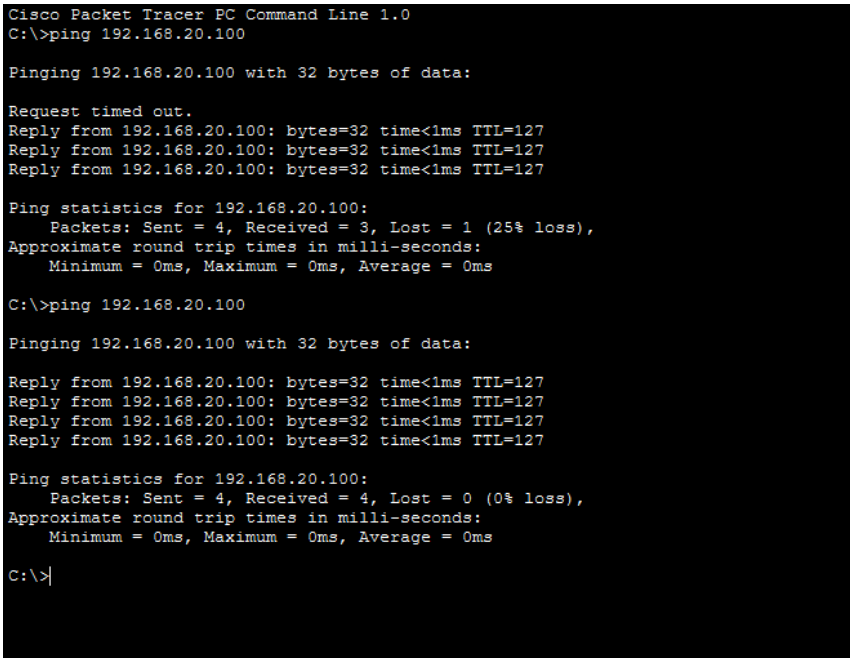
Router#
```

Testing and Results

The following tests were completed successfully:

- All four PCs received their network settings through DHCP.
- Admin and IT devices communicated across the two subnets.
- intranet.office resolved to 192.168.10.2.
- The internal website loaded from both Admin and IT PCs.





Troubleshooting

Initially, the server-based DHCP setup did not provide the correct gateway and DNS settings to the PCs. DHCP was moved to the router, and the DHCP pools were configured for both subnets. After this change, all four PCs received the expected settings and network connectivity was verified.

Conclusion

This Packet Tracer simulation demonstrated how a small office network can use separate departmental subnets, router-based DHCP, inter-network routing, DNS name resolution, and an internal web service. The project provided hands-on practice with network configuration, connectivity testing, and troubleshooting.